

External side effects and recovery

Project: Grounded Learning Service

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Timeouts retries and cancellation

Without opening the old lesson, explain one key idea from Timeouts retries and cancellation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Planning and action selection

Without opening the old lesson, explain one key idea from Planning and action selection. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Failure budgets and service objectives

Without opening the old lesson, explain one key idea from Failure budgets and service objectives. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for External side effects and recovery. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for External side effects and recovery. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

A payment-like classroom token transfer has unknown status after network loss. Should the mock repeat immediately?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

A booking is committed in storage, but response delivery times out. Caller asks to repeat. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

A payment-like classroom token transfer has unknown status after network loss. Should the mock repeat immediately?

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled external side effects and recovery evidence sheet to Grounded Learning Service. Show how you can explain the mechanism and terminology of external side effects and recovery. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week’s record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: External side effects and recovery

Recover an uncertain print job. A badge printer loses its reply after receiving a print request.

Job key J6 identifies one badge. The printer supports status lookup by key; missing acknowledgement leaves outcome unknown.

External side effects and recovery

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Record uncertainty

Sending occurred, but printing completion is not yet known.



2. Query J6

Ask for the existing job's recorded state.



3. Recover from status

If complete, return that result without issuing another print job.

CHANGE ONE THING

Status reliably reports J6 absent.

Explain the changed result using the matching lesson.

Distinguish a missing printer acknowledgement from confirmed failure. What should happen if J6 is complete, and what evidence changes when J6 is reliably reported absent?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

13. Project portfolio milestone

Project: Grounded Learning Service

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Recover an uncertain print job. A badge printer loses its reply after receiving a print request.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? If complete, return that result without issuing another print job.

1. Record uncertainty
2. Recover from status
3. Query J6

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Recover from status If complete, return that result without issuing another print job. A lost response does not show whether an external effect occurred. Recovery distinguishes unknown outcome from a confirmed failed action.

Practice 2: explain the deciding evidence

Changed-case practice: Status reliably reports J6 absent. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. A policy-approved resend using the same deduplication key may be considered.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: A policy-approved resend using the same deduplication key may be considered. Confirmed absence supplies different evidence from silence.

Repair a missing building block

Review Timeouts retries and cancellation (ai-6/16). Goal: Explain the mechanism and terminology of timeouts retries and cancellation. Short practice: Calculate waiting time for a one-second timeout followed by a 0.4-second success. What changes if cancellation arrives after the first attempt but before the retry?. Return to this week and explain what you changed.

Review Planning and action selection (ai-6/14). Goal: Explain the mechanism and terminology of planning and action selection. Short practice: Choose a permitted plan when no approved tower-height card exists. What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?. Return to this week and explain what you changed.

Review Failure budgets and service objectives (ai-6/11). Goal: Explain the mechanism and terminology of failure budgets and service objectives. Short practice: Calculate the maximum allowed failures among 100 requests. Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
